INTEGRATION DOCUMENTATION COLLECTION SERVICE (API)

V1.2 – 2025

I. OVERVIEW

I.1. DEFINITIONS

Terms	Description
Pay2Pay	Pay2Pay Payment Gateway
Merchant	Partners using the services of Pay2Pay
Bank	The bank where the partner's customers want to receive money
HTTPS	Secure Hypertext Transfer Protocol
TLS	Transfer Layer Security
Signature	To ensure data transmission between Merchant and Pay2Pay is secure and accurate, all data transmitted between the two parties will be encrypted and digitally signed to ensure data constraints.

I.2. DOCUMENT PURPOSE

The document describes the detailed steps necessary for partners to integrate into the Pay2Pay Payment Gateway system.

I.3. OBJECT

This document is used for partner developers to integrate into the Pay2Pay Payment Gateway system.

II. COMMUNICATION MODEL

The following figure describes an overview of the communication model between the Merchant's system and the Pay2Pay system.

- Pay2Pay uses HTTPS/TLS1.2 connection standard

- Every request sent from merchant must be authenticated by the JSON Web Token solution.

- Pay2Pay provides APIs according to Restful Webservice standards, transmitting and receiving data in JSON format.

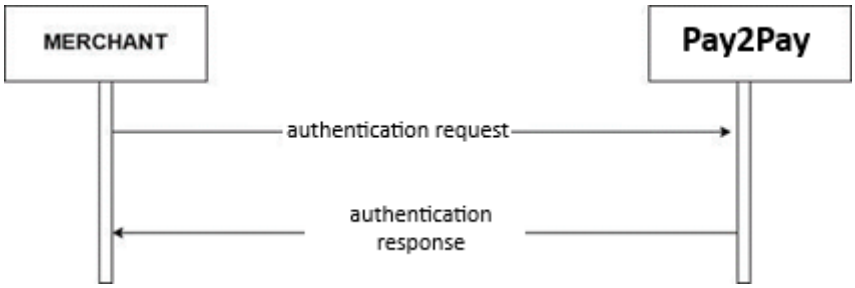
- To ensure data transmission and reception between the two parties is secure and authentic, all data will be encrypted and digitally signed.

III. JSON WEBTOKEN AUTHORIZATION & DIGITAL SIGNATURE

III.1. JSON WEBTOKEN AUTHORIZATION

All requests from merchant to Pay2Pay system need to be authorized to ensure security.

Processing Flow:



Request:

#	Parameters	Data Type	Condition	Description
Header				
1	p-request-id	String	Mandatory	A unique code corresponding to a request sent from the merchant system
2	p-request-time	String	Mandatory	The time of request. Format YYYYMMDDHHMMSS
3	p-tenant	String	Mandatory	<TENANT INFO>. This information is sent along with username and password
4	p-signature	String	Mandatory	The signature is generated by the RSA-2048 algorithm
Body				
1	username	String	Mandatory	Login account name
2	password	String	Mandatory	Password hashed according to the SHA256 algorithm including username.

Respose:

#	Parameters	Data Type	Description
1	code	String	Result code
2	message	String	Message
3	accessToken	String	Used to access other APIs
4	refreshToken	String	Used to get new accessToken
5	expireIn	Number	Expired time of accessToken(seconds)
6	refreshExpiresIn	Number	Expired time of refreshToken(tính theo giây).
7	Profile	Object	Login profile if available

III.2. DIGITAL SIGNATURE

All data transmitted and received between the two parties will be encrypted and digitally signed.

ALGORITHM:

- Signature is created based on RSA-2048 algorithm.

- Merchant creates RSA-2048 key pair - including private key and public key - through online tools and sends Public key to Pay2Pay.

- Private key for merchant to create signatures.

- Public key provided by the merchant to Pay2Pay to verify the merchant's signature.

HOW TO CREATE A SIGNATURE:

Step 1: Filter and sort header parameters alphabetically (authorization, verification & fields prefixed with p-) and concatenate header values.

Step 2: Take the header values and combine them with the request body to get the result of the string to be signed

Step 3: Signature = SHA256withRSA(String to be signed in Step 2)

FOR EXAMPLE:

SHA256withRSA(RSA private key, “5d599b0b-c728-4c18-9748-17f38b4cc9a620240521115518MERCHANT-WEB{"username":"0982569888","password":"YTEyYWQxYjc3MGM0MTg4ZWQ5ZDQ3MTZlODNjYTJhZWRmOTFiMjliNjJhNzY0YjI4MzI3NWY5OGUyZWE2NWMwOA==\"}”)

RESULT OF SIGNATURE CREATION:

UNJDNul1ErQrci6RrNXfWFDnJUAXKSAOkZAejFDYCzqnm8jqZt5cns2lS0C1MVCbr1gm9mz5+OiJWJJPlxxsd5hn9G0izWhgvZ4MbLpYzzHFRmarLvqwsfSsJSAk5iaPlMMHHUy4hu5dr1XURRQjy42xJR6h/5vkyPLxzkjMEpeWYCd/d9nQKfBi7vcp+zAPwH+7vNzyT/37tqmT5sl/KYvoLPS47nOkJHuhOct8LmjQJmQBPGUiwr5U048/ZiqQmAb9Ki62/Dy8Vh3CWPybTKKLSfIXIvOglwNHhU0y7imLE6nVO5puBigLVg2BjX9M9MQzIr/42cD9x4nNMSs8Kw==

Refer link to check: https://static.pay2pay.vn/tools/RSA2048_Tools

IV. SYSTEM INFORMATION

IV.1. DOMAIN

Merchant will use the following domain for all Pay2Pay Payment Gateway APIs.

Environment	URL
Test	https://uat-api.pay2pay.vn
Production	https://api.pay2pay.vn

IV.2. MERCHANT ADMIN WEB INFORMATION

Merchant will be provided with an account to login the Merchant Administration Page built by Pay2Pay to track payment transactions.

Environment	URL
Test	https://uat-merchant.pay2pay.vn
Production	https://merchant.pay2pay.vn

Merchant Administration Page login account information will be provided when the two parties proceed with the integration.

IV.3. INFORMATION MERCHANT NEEDS TO PROVIDE

Merchant needs to provide the following information to Pay2Pay

Info	Condtion	Description
IP address	Mandatory	List of IP addresses wil be used to send requests to Pay2Pay. - Required on Production environment - Optional on Test environment.
ipn_url	Mandatory	URL to receive transaction results from Pay2Pay, sent immediately after confirmation from the bank.
redirect_url	Optional	URL for Pay2Pay to transfer customers to the merchant site, after the customer completes the transfer on the Pay2Pay site.
Public Key	Mandatory	The public key the merchant provides to Pay2Pay allows Pay2Pay to verify the merchant's signature when sending a request to Pay2Pay.

V. LIST OF API

V.1. LOGIN

API to get access token

URL: https://{DOMAIN}/auth-service/api/v1.0/user/login
Method: POST
Request

#	Parameters	Data Type	Condition	Description
Header				
1	p-request-id	String	Mandatory	A unique code corresponding to a request sent from the merchant system
2	p-request-time	String	Mandatory	Time to request API (Unix time from 00:00:00 January 1, 1970). Format YYYYMMDDHHMMSS Note: It should be convert to UTC + 7
3	p-tenant	String	Mandatory	<TENANT INFO>. This information is sent along with username and password
4	p-signature	String	Mandatory	The signature is generated by the RSA-2048 algorithm
Body				
1	username	String	Mandatory	Login account name
2	password	String	Mandatory	Password hashed according to the SHA256 algorithm including username. Password value = SHA256(username + raw password). Ex: username: user password: 123456 Password value = SHA256(user123456) **Refer to this link** https://static.pay2pay.vn/tools/sha256_hex_base64

Respose

#	Parameters	Data Type	Description
1	code	String	Result code
2	message	String	Message
3	accessToken	String	Used to access other APIs
4	refreshToken	String	Used to get new accessToken
5	expireIn	Number	Expired time of accessToken(seconds)
6	refreshExpiresIn	Number	Expired time of refreshToken(tính theo giây).
7	Profile	Object	Login profile if available

EXAMPLE:
Request

```
curl --location 'https://{DOMAIN}/auth-service/api/v1.0/user/login' \
--header 'p-request-id: e3d072af-adf6-4d98-9188-5ba696fe7381' \
--header 'p-request-time: 20240520093724' \
--header 'p-tenant: MERCHANT-WEB' \
```

--header 'p-signature:
HLA0/lgWUpDVw+aGXcsNL8yZpgWl7zU/xhRanDi87K5Oj52+ZQJv4XOl4n66rYYyhyOxKw3Puw5hYZdDD
P7eOsvFO3b0kUu2LONQxq+A4vIpyJETjaqGPgVOr2mPzOzgXjiAco44dXRQFPpfTKNOrJy+IcRMgMgDJB9Z
gI6dp2yMrIw5Mp8tFPeTEwYbpFTBFX8XDlDfCPtBLydvTijWuExny+gmX4LgkuzH1Cz89cMn4+rMCudsJfeW
Wd09iVQ1FLQkH9XNOGXe3HEApv+gSgaiLm/Hb3H9q5hsDuzVRZTZaamd38YxlANoYTy7lY8tPgtc4PWwA
VJ15V1Dv4lU6w==' \
--header 'Content-Type: application/json' \
--data '{
"username": "098256988",
"password":
"YTEyYWQxYjc3MGM0MTg4ZWQ5ZDQ3MTZlODNjYTJhZWRmOTFiMjliNjJhNzY0YjI4MzI3NWY5OGU
yZWE2NWMwOA=="
}'

Response

{
"code": "SUCCESS",
"message": "Thành công.",
"data": {
"accessToken":
"eyJhbGciOiJSUzI1NiIsInR5cCIgOiAiSldUIiwia2lkIiA6ICJFdFZjdmtDbUFYcWZWeU1TVjhhOThscnhUV1lJU
lJYRFprTER6UVZQekdBIn0.eyJleHAiOjE3MTYyNTc1MzAsImlhdCI6MTcxNjI1NjkzMCwianRpIjoiY2MzNzF
hYjUtOGM4YS00NjU0LWE2MzgtODNhN2YyNGY1MWUxIiwiaXNzIjoiaHR0cHM6Ly9wcGlkLnVhdC5wYX
lwYXkuY29tLnZuL3JlYWxtcy93YWxsZXQiLCJhdWQiOiJhY2NvdW50Iiwic3ViIjoiOWVhMmY3OGItNzg2Zi
00MTZlLWFlZGEtYWEyMGUzNjY0ZGQ3IiwidHlwIjoiQmVhcmVyIiwiYXpwIjoicGF5cGF5LXdhbGxldCIsIn
Nlc3Npb25fc3RhdGUiOiI5M2FhODFhZi0xNjlmLTQ5NDUtOTc4MS04ZTFkNjQyYjgwZjYiLCJhY3IiOiIxIiwi
YWxsb3dlZC1vcmlnaW5zIjpbIi8qIl0sInJlYWxtX2FjY2VzcyI6eyJyb2xlcyI6WyJvZmZsaW5lX2FjY2VzcyIsInVt
YV9hdXRob3JpemF0aW9uIiwiZGVmYXVsdC1yb2xlcy13YWxsZXQiXX0sInJlc291cmNlX2FjY2VzcyI6eyJhY
2NvdW50Ijp7InJvbGVzIjpbIm1hbmFnZS1hY2NvdW50IiwibWFuYWdlLWFjY291bnQtbGlua3MiLCJ2aWV3L
XByb2ZpbGUiXX19LCJzY29wZSI6IndhbGxldC1jbGllbnQtc2NvcGUgZW1haWwgcHJvZmlsZSIsInNpZCI6Ijk
zYWE4MWFmLTE2OWYtNDk0NS05NzgxLThlMWQ2NDJiODBmNiIsImVtYWlsX3ZlcmlmaWVkIjpmYWxz
ZSwibWVyY2hhbnRJZCI6IlBQMDAwMDA3MCIsIm5hbWUiOiJkZWZhdWx0IGRlZmF1bHQiLCJncm91cHM
iOlsiL21hIiwiL21lcmNoYW50IiwiL29wcyJdLCJwcmVmZXJyZWRfdXNlcm5hbWUiOiIwNzgzMTgwODc4Iiwi
Z2l2ZW5fbmFtZSI6ImRlZmF1bHQiLCJmYW1pbHlfbmFtZSI6ImRlZmF1bHQifQ.MZsB4742DAqNIQMy4op9
TdwrhUyAM6T-L9WRmdXcD8FUMGRXJwuW-6XntQRyVZMKlfHNS_C-EIP44rAYV-
4BMogLXdBWIPt1EVDDnEKHlGQU3RmUPnevFQ-
NVyuCl6sfkNlwjB_u5MuOxZhJrT7rKYrEVLiKpuzUGK5fz6dCk0foP5xQXHuz1t4edw2tK2vzFZYrKCY00wiU
P_yKF7uAduf0le7TFLR_tkerEKhyYaQyPxdSaPggzxJF3Utfc_YB4mgQsXc0JkkGxHPfFPQA9RKZbc5GE9x_V
Zpd2r1nwq_I0TX5SEVJUUT72dCiCZktJpmnFuw7U0Me7FFHUh0qpQ",
"refreshToken":
"eyJhbGciOiJIUzI1NiIsInR5cCIgOiAiSldUIiwia2lkIiA6ICI3MTljOTI0ZS1jY2U3LTQ5NTMtODYzZi1kZjU0NT
kxNDJkZWIifQ.eyJleHAiOjE3MTYyNTg3MzAsImlhdCI6MTcxNjI1NjkzMCwianRpIjoiZTYzMTMxOTItNDkz
MC00M2ViLWFkYjYtZjA2NjhlZDk5ZjgzIiwiaXNzIjoiaHR0cHM6Ly9wcGlkLnVhdC5wYXlwYXkuY29tLnZu
L3JlYWxtcy93YWxsZXQiLCJhdWQiOiJodHRwczovL3BwaWQudWF0LnBheXBheS5jb20udm4vcmVhbG1zL3
dhbGxldCIsInN1YiI6IjllYTJmNzhiLTc4NmYtNDE2ZS1hZWRhLWFhMjBlMzY2NGRkNyIsInR5cCI6IlJlZnJlc
2giLCJhenAiOiJwYXlwYXktd2FsbGV0Iiwic2Vzc2lvbl9zdGF0ZSI6IjkzYWE4MWFmLTE2OWYtNDk0NS05N
zgxLThlMWQ2NDJiODBmNiIsInNjb3BlIjoid2FsbGV0LWNsaWVudC1zY29wZSBlbWFpbCBwcm9maWxlIiwi
c2lkIjoiOTNhYTgxYWYtMTY5Zi00OTQ1LTk3ODEtOGUxZDY0MmI4MGY2In0.KSaklQgGp3Hz4oW6ConK
RUC1Bd0eBglPSNor0YwqCj4",
"expireIn": "600",
"refreshExpiresIn": "1800",
"profile": null
}
}

V.2. REFRESH TOKEN

API to get new access token

URL: https://{DOMAIN}/auth-service/api/v1.0/user/refresh

Method: POST

Request

#	Parameters	Data Type	Condition	Description
Header				
1	p-request-id	String	Mandatory	A unique code corresponding to a request sent from the merchant system

2	p-request-time	String	Mandatory	Time to request API (Unix time from 00:00:00 January 1, 1970). Format YYYYMMDDHHMMSS Note: It should be convert to UTC + 7
3	p-tenant	String	Mandatory	<TENANT INFO>. This information is sent along with username and password
4	p-signature	String	Mandatory	The signature is generated by the RSA-2048 algorithm
Body				
1	refreshToken	String	Mandatory	Used to get new access token

Response

#	Parameters	Data Type	Description
1	code	String	Result code
2	message	String	Message
3	accessToken	String	Used to access other APIs
4	refreshToken	String	Used to get new accessToken
5	expireIn	Number	Expired time of accessToken(seconds)
6	refreshExpiresIn	Number	Expired time of refreshToken(tính theo giây).

EXAMPLE:

Request

```
curl --location 'https://{DOMAIN}/auth-service/api/v1.0/user/refresh' \
--header 'p-request-id: 237b185a-42ae-4951-9032-493f6b8e30e5' \
--header 'p-request-time: 20240521090414' \
--header 'p-tenant: MERCHANT-WEB' \
--header 'Authorization: Bearer
eyJhbGciOiJSUzI1NiIsInR5cCIgOiAiSldUIiwia2lkIiA6ICJFdFZjdmtDbUFYcWZWeU1TVjhhOThscnhUV1lJUl
JYRFprTER6UVZQekdBIn0.eyJleHAiOjE3MTYyNTc1MzAsImlhdCI6MTcxNjI1NjkzMCwianRpIjoiY2MzNzF
hYjUtOGM4YS00NjU0LWE2MzgtODNhN2YyNGY1MWUxIiwiaXNzIjoiaHR0cHM6Ly9wcGlkLnVhdC5wYX
lwYXkuY29tLnZuL3JlYWxtcy93YWxsZXQiLCJhdWQiOiJhY2NvdW50Iiwic3ViIjoiOWVhMmY3OGItNzg2Zi
00MTZlLWFlZGEtYWEyMGUzNjY0ZGQ3IiwidHlwIjoiQmVhcmVyIiwiYXpwIjoicGF5cGF5LXdhbGxldCIsIn
Nlc3Npb25fc3RhdGUiOiI5M2FhODFhZi0xNjlmLTQ5NDUtOTc4MS04ZTFkNjQyYjgwZjYiLCJhY3IiOiIxIiwi
YWxsb3dlZC1vcmlnaW5zIjpbIi8qIl0sInJlYWxtX2FjY2VzcyI6eyJyb2xlcyI6WyJvZmZsaW5lX2FjY2VzcyIsInVt
YV9hdXRob3JpemF0aW9uIiwiZGVmYXVsdC1yb2xlcy13YWxsZXQiXX0sInJlc291cmNlX2FjY2VzcyI6eyJhY
2NvdW50Ijp7InJvbGVzIjpbIm1hbmFnZS1hY2NvdW50IiwibWFuYWdlLWFjY291bnQtbGlua3MiLCJ2aWV3L
XByb2ZpbGUiXX19LCJzY29wZSI6IndhbGxldC1jbGllbnQtc2NvcGUgZW1haWwgcHJvZmlsZSIsInNpZCI6Ijk
zYWE4MWFmLTE2OWYtNDk0NS05NzgxLThlMWQ2NDJiODBmNiIsImVtYWlsX3ZlcmlmaWVkIjpmYWxz
ZSwibWVyY2hhbnRJZCI6IlBQMDAwMDA3MCIsIm5hbWUiOiJkZWZhdWx0IGRlZmF1bHQiLCJncm91cHM
iOlsiL21hIiwiL21lcmNoYW50IiwiL29wcyJdLCJwcmVmZXJyZWRfdXNlcm5hbWUiOiIwNzgzMTgwODc4Iiwi
Z2l2ZW5fbmFtZSI6ImRlZmF1bHQiLCJmYW1pbHlfbmFtZSI6ImRlZmF1bHQifQ.MZsB4742DAqNIQMy4op9
TdwrhUyAM6T-L9WRmdXcD8FUMGRXJwuW-6XntQRyVZMKlfHNS_C-EIP44rAYV-
4BMogLXdBWIPt1EVDDnEKHlGQU3RmUPnevFQ-
NVyuCl6sfkNlwjB_u5MuOxZhJrT7rKYrEVLiKpuzUGK5fz6dCk0foP5xQXHuz1t4edw2tK2vzFZYrKCY00wiU
P_yKF7uAduf0le7TFLR_tkerEKhyYaQyPxdSaPggzxJF3Utfc_YB4mgQsXc0JkkGxHPfFPQA9RKZbc5GE9x_V
Zpd2r1nwq_I0TX5SEVJUUT72dCiCZktJpmnFuw7U0Me7FFHUh0qpQ' \
--header 'p-signature:
ZvNkKQyDumByQd/JReUOQrY3cIZm8CHTLCThMw65U4QWg7LzkSha8EtOrlDv1PjLwRCbbzpckJBIjhlBhv
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UWjvDoI8UDSKMWQVpELW4TEo+gsHq4CLwv9A68sh05LOKk3o9FJK4hm0vY5kZwZQYb9qjFsv9tk5gNm
alabAiBl0n8bwyS7pZVNEJZEimSLGMXoMwhwgf9olEl/aGsT249Ce/0/QzJk8DVx3slu3QL9Zl6En6SX4YvnqV
rW3EdrI1IFSLks8hA==' \
--header 'Content-Type: application/json' \
--data
'{"refreshToken":"eyJhbGciOiJIUzI1NiIsInR5cCIgOiAiSldUIiwia2lkIiA6ICI3MTljOTI0ZS1jY2U3LTQ5NTMtO
```

DYzZi1kZjU0NTkxNDJkZWIifQ.eyJleHAiOjE3MTYyNTg3MzAsImlhdCI6MTcxNjI1NjkzMCwianRpIjoiZTYz
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wYXkuY29tLnZuL3JlYWxtcy93YWxsZXQiLCJhdWQiOiJodHRwczovL3BwaWQudWF0LnBheXBheS5jb20ud
m4vcmVhbG1zL3dhbGxldCIsInN1YiI6IjllYTJmNzhiLTc4NmYtNDE2ZS1hZWRhLWFhMjBlMzY2NGRkNyIs
InR5cCI6IlJlZnJlc2giLCJhenAiOiJwYXlwYXktd2FsbGV0Iiwic2Vzc2lvbl9zdGF0ZSI6IjkzYWE4MWFmLTE2O
WYtNDk0NS05NzgxLThlMWQ2NDJiODBmNiIsInNjb3BlIjoid2FsbGV0LWNsaWVudC1zY29wZSBlbWFpbC
Bwcm9maWxlIiwic2lkIjoiOTNhYTgxYWYtMTY5Zi00OTQ1LTk3ODEtOGUxZDY0MmI4MGY2In0.KSaklQg
Gp3Hz4oW6ConKRUC1Bd0eBglPSNor0YwqCj4"}"

Response

{
"code": "SUCCESS",
"message": "Thành công.",
"data": {
"accessToken":
"eyJhbGciOiJSUzI1NiIsInR5cCIgOiAiSldUIiwia2lkIiA6ICJFdFZjdmtDbUFYcWZWeU1TVjhhOThscnhUV1lJU
lJYRFprTER6UVZQekdBIn0.eyJleHAiOjE3MTYyNzQzMjUsImlhdCI6MTcxNjI3MzcyNSwianRpIjoiMTg3NG
Y0YTEtNGVlOS00YzcxLWI4MWYtZWU2NzNjMTA3MTdiIiwiaXNzIjoiaHR0cHM6Ly9wcGlkLnVhdC5wYX
lwYXkuY29tLnZuL3JlYWxtcy93YWxsZXQiLCJhdWQiOiJhY2NvdW50Iiwic3ViIjoiNjQ2YmQ5NGItMTI0MC
00NTNkLWI0NTYtZjY1OWRkN2MzOTFjIiwidHlwIjoiQmVhcmVyIiwiYXpwIjoicGF5cGF5LXdhbGxldCIsIn
Nlc3Npb25fc3RhdGUiOiJhNDlhMzQ3NC05Y2Y0LTQwZmEtYTY2Ny0zZGE1ZjlkMzAyM2MiLCJhY3IiOiIxI
iwiYWxsb3dlZC1vcmlnaW5zIjpbIi8qIl0sInJlYWxtX2FjY2VzcyI6eyJyb2xlcyI6WyJvZmZsaW5lX2FjY2VzcyIsI
nVtYV9hdXRob3JpemF0aW9uIiwiZGVmYXVsdC1yb2xlcy13YWxsZXQiXX0sInJlc291cmNlX2FjY2VzcyI6ey
JhY2NvdW50Ijp7InJvbGVzIjpbIm1hbmFnZS1hY2NvdW50IiwibWFuYWdlLWFjY291bnQtbGlua3MiLCJ2aW
V3LXByb2ZpbGUiXX19LCJzY29wZSI6IndhbGxldC1jbGllbnQtc2NvcGUgZW1haWwgcHJvZmlsZSIsInNpZCI
6ImE0OWEzNDc0LTljZjQtNDBmYS1hNjY3LTNkYTVmOWQzMDIzYyIsImVtYWlsX3ZlcmlmaWVkIjpmY
WxzZSwibmFtZSI6ImRlZmF1bHQgZGVmYXVsdCIsImdyb3VwcyI6WyIvbWEiLCIvbWVyY2hhbnQiLCIvb3B
zIiwiL3NidiIsIi93YWxsZXQiXSwicHJlZmVycmVkX3VzZXJuYW1lIjoiMDk4MjU2OTg4NyIsImdpdmVuX25h
bWUiOiJkZWZhdWx0IiwiZmFtaWx5X25hbWUiOiJkZWZhdWx0In0.rcfDum_bLNoWYXqXQuWgIiaczSKizv
G177qpS19XDGmJZw5OMrJGCIDapET0URkYF4iOXjhieERHnhTB_NaEjDkObAt6h6Y4wuZsTX3TpDGbekz
glPQxO_8ananHZ7G_X-iNjz4vuape-UwLB5-81-NunNL7wm0GP3kZ_H3qAV-
sS3JDqnT5SwOIFRrK2Trn2LWMRHb78HwIln5py0djW5mltgOS-TpBjAZ9erEhwVXHSbl414j-
FnyTR2NPVrUVjbQXsO9cyYiBJx_HF2QtRu453pQTctNxm8UbtyTo5AXThjf21UaTsmtFrfdu1dtAqmH5Nj_Q8
9I_rWb-rtehrg",
"refreshToken":
"eyJhbGciOiJIUzI1NiIsInR5cCIgOiAiSldUIiwia2lkIiA6ICI3MTljOTI0ZS1jY2U3LTQ5NTMtODYzZi1kZjU0NT
kxNDJkZWIifQ.eyJleHAiOjE3MTYyNzU1MjUsImlhdCI6MTcxNjI3MzcyNSwianRpIjoiNzdlYmVmYjQtOWI1
OC00NTFmLWI4MGYtNGY1Y2JhNWU0NzQ5IiwiaXNzIjoiaHR0cHM6Ly9wcGlkLnVhdC5wYXlwYXkuY29
tLnZuL3JlYWxtcy93YWxsZXQiLCJhdWQiOiJodHRwczovL3BwaWQudWF0LnBheXBheS5jb20udm4vcmVhb
G1zL3dhbGxldCIsInN1YiI6IjY0NmJkOTRiLTEyNDAtNDUzZC1iNDU2LWY2NTlkZDdjMzkxYyIsInR5cCI6Il
JlZnJlc2giLCJhenAiOiJwYXlwYXktd2FsbGV0Iiwic2Vzc2lvbl9zdGF0ZSI6ImE0OWEzNDc0LTljZjQtNDBmYS
1hNjY3LTNkYTVmOWQzMDIzYyIsInNjb3BlIjoid2FsbGV0LWNsaWVudC1zY29wZSBlbWFpbCBwcm9maW
xlIiwic2lkIjoiYTQ5YTM0NzQtOWNmNC00MGZhLWE2NjctM2RhNWY5ZDMwMjNjIn0.iLjLIf815LwqbSF2
GVqD43y7WVyMR-UXX-IfLN9NaBk",
"expireIn": "600",
"refreshExpiresIn": "1800"
}
}

V.3. PAYMENT METHOD

API to get list of payment methods

URL: https://{DOMAIN}/bank-gateway-service/mch/api/v1.0/payment_method

Method: GET

Request

#	Parameters	Data Type	Condition	Description
Header				
1	p-request-id	String	Mandatory	A unique code corresponding to a request sent from the merchant system
2	p-request-time	String	Mandatory	Time to request API (Unix time from 00:00:00 January 1, 1970). Format YYYYMMDDHHMMSS Note: It should be convert to UTC + 7

3	p-tenant	String	Mandatory	<TENANT INFO>. This information is sent along with username and password
4	p-signature	String	Mandatory	The signature is generated by the RSA-2048 algorithm
#	**Parameters**	**Data Type**	**Condition**	**Description**

Response

#	Parameters	Data Type	Description
1	code	String	Result code
2	message	String	Message
	paymentMethodInfo	List	List of payment methods
3	code	String	Payment method code
4	status	boolean	Status of payment method
5	logoPath	String	Logo of payment method
6	backGround	String	Background of payment method
7	type	int	Type of payment method
8	name.vi	String	Name in Vietnamese of payment method
9	name.en	String	Name in English of payment method
10	issuers.code	String	Issuer Code
11	issuers.bin	Number	Bin Code
12	issuers.name.vi	String	Issuer Name in Vietnamese
13	issuers.name.en	String	Issuer Name in English
14	issuers.logoPath	String	Logo of issuer
15	issuers.status	boolean	Status of Issuer
16	issuers.paymentType	String	Payment Type
17	issuers.type	String	Type of issuer
18	issuers.cardRegex	String	Issuer Card format
19	issuers.accountRegex	String	Issuer Account format
20	issuers.order	Number	Display Order
21	issuers.currency.code	String	Currency Code
22	issuers.currency.minimum	Number	Minimum amount
23	issuers.currency.maximum	Number	Maximum amount

EXAMPLE:

Request

```
curl --location 'https://{DOMAIN}/bank-gateway-service/mch/api/v1.0/payment_method' \
--header 'p-request-id: bc76ce46-3ab1-48b5-8d30-78f4823a39d8' \
--header 'p-request-time: 20240523095427' \
--header 'p-tenant: MERCHANT-WEB' \
```

--header 'Authorization: Bearer
eyJhbGciOiJSUzI1NiIsInR5cCIgOiAiSldUIiwia2lkIiA6ICJFdFZjdmtDbUFYcWZWeU1TVjhhOThscnhUV1lJUl
JYRFprTER6UVZQekdBIn0.eyJleHAiOjE3MTY0MzM0NDAsImlhdCI6MTcxNjQzMjg0MCwianRpIjoiZTgyM
WNhOGUtNjdjZC00MDQwLTkwODQtZDU1OTcyYTQzNGQwIiwiaXNzIjoiaHR0cHM6Ly9wcGlkLnVhdC5w
YXlwYXkuY29tLnZuL3JlYWxtcy93YWxsZXQiLCJhdWQiOiJhY2NvdW50Iiwic3ViIjoiNjQ2YmQ5NGItMTI0
MC00NTNkLWI0NTYtZjY1OWRkN2MzOTFjIiwidHlwIjoiQmVhcmVyIiwiYXpwIjoicGF5cGF5LXdhbGxldCI
sInNlc3Npb25fc3RhdGUiOiI5ZmY1OWUxYi1iMzFhLTQyNTYtYjk3NC1hODU5ZDU3ODBiNWYiLCJhY3Ii
OiIxIiwiYWxsb3dlZC1vcmlnaW5zIjpbIi8qIl0sInJlYWxtX2FjY2VzcyI6eyJyb2xlcyI6WyJvZmZsaW5lX2FjY2Vz
cyIsInVtYV9hdXRob3JpemF0aW9uIiwiZGVmYXVsdC1yb2xlcy13YWxsZXQiXX0sInJlc291cmNlX2FjY2Vzcy
I6eyJhY2NvdW50Ijp7InJvbGVzIjpbIm1hbmFnZS1hY2NvdW50IiwibWFuYWdlLWFjY291bnQtbGlua3MiLCJ2
aWV3LXByb2ZpbGUiXX19LCJzY29wZSI6IndhbGxldC1jbGllbnQtc2NvcGUgZW1haWwgcHJvZmlsZSIsInNp
ZCI6IjlmZjU5ZTFiLWIzMWEtNDI1Ni1iOTc0LWE4NTlkNTc4MGI1ZiIsImVtYWlsX3ZlcmlmaWVkIjpmYWx
zZSwibWVyY2hhbnRJZCI6IlBQMDAwMDAwMSIsIm5hbWUiOiJkZWZhdWx0IGRlZmF1bHQiLCJncm91cH
MiOlsiL21hIiwiL21lcmNoYW50IiwiL29wcyIsIi9zYnYiLCIvd2FsbGV0Il0sInByZWZlcnJlZF91c2VybmFtZSI6Ij
A5ODI1Njk4ODciLCJnaXZlbl9uYW1lIjoiZGVmYXVsdCIsImZhbWlseV9uYW1lIjoiZGVmYXVsdCJ9.R60niC
8SR4sYeZxYf50vKVaVABB7dA85wT4WhoUqeyd-
RaquHmzC_jiDmVMSmkphMmrSQuIM4z0CZgeQTIci2YeaWlQrflqMfc05ysfzeR-
wLlvTRz_ZzvEao15uUKGToOLhPi1z9YUzqoU_ipJPtPaS6reLMo1N0UT2Mg5VyX7J9nGn4AdySeJHFtOnbpD
_bFdGsCRui053Ecc_nnm0pytWHrGDahoT4QPN6Zny9IjROkCNytuEvSN5G2UbyIeshmvnG3uWb_VZNSchCF
8T9sab1TCl_s7EjZ98cOgq77zNHHeknnxgT5EhhUL0Xkg3lqwMBuDGsT46OSfvEJ_gFg' \
--header 'p-signature:
fXIEIv7ydhfcD8wOgKo0jr2tBdya7O7fbKj359px4HVoUNILXagj+Lcr2oVDRy5shkkT+71Uggg3SRFQP2q+AsV
dX/DSm0AnLvuQZ5Gn/n8uRasq3sxqZ8dQZA9bvg6HADsqohnJKPvfJiLaVTnPQMpWTAAao0rot9MDFkG7d
Rpyo5Sqz45BkD6VMeGhtZjiD5s8jsi8IJA8cCHwEwu7QDqji32V82MrumoU5v4PKMzCXEVff6/NptmIhhxcioG
A/j/612XdK6Gll7fN9Ow1plS90uPTAXlirFTVv9YtIfT47lZJnNUhb3vUF21jt3fXe8D3O3fwVHCvXgeHeHrfVQ
=='

Response

{
 "code": "SUCCESS",
 "message": "Thành công.",
 "data": {
  "paymentMethodInfo": [
   {
    "code": "DOMESTIC",
    "name": {
     "vi": "Tài khoản ngân hàng\nThẻ ATM nội địa",
     "en": "DOMESTIC"
    },
    "issuers": [
     {
      "code": "TPB",
      "bin": "970423&970402",
      "name": {
       "vi": "Ngân hàng TMCP Tiên Phong",
       "en": "Tien Phong Commercial Joint Stock Bank"
      },
      "logoPath": "https://static.pay2pay.vn.vn/bank-logo/11-tpbank.png",
      "status": true,
      "paymentType": "CARD",
      "type": "BANK",
      "cardRegex": null,
      "accountRegex": null,
      "order": 34,
      "currency": {
       "code": "VND",
       "minimum": 10000,
       "maximum": 20000000
      }
     },
     {
      "code": "BIDV",
      "bin": "970418",
      "name": {
       "vi": "Ngân hàng TMCP Đầu tư và phát triển Việt Nam",
       "en": "Bank for Investment and Development of Vietnam"
      },
      "logoPath": "https://static.pay2pay.vn.vn/bank-logo/5-bidv.png",

```
            "status": true,
            "paymentType": "CARD",
            "type": "BANK",
            "cardRegex": null,
            "accountRegex": null,
            "order": 1,
            "currency": {
              "code": "VND",
              "minimum": 10000,
              "maximum": 20000000
            }
          }
        ],
        "status": true,
        "logoPath": "https://static.pay2pay.vn.vn/payment/logo/domestic.jpg",
        "backGround": null,
        "type": "FORM"
      },
      {
        "code": "QR",
        "name": {
          "vi": "Ví điện tử Pay2Pay",
          "en": "QR"
        },
        "issuers": [
          {
            "code": "PAY2PAY",
            "bin": "970416",
            "name": {
              "vi": "Ví điện tử Pay2Pay",
              "en": "Pay2Pay ewallet"
            },
            "logoPath": "",
            "status": true,
            "paymentType": "CARD_ACCOUNT",
            "type": "BANK",
            "cardRegex": null,
            "accountRegex": null,
            "order": 1,
            "currency": {
              "code": "VND",
              "minimum": 10000,
              "maximum": 1000000
            }
          }
        ],
        "status": true,
        "logoPath": "https://static.pay2pay.vn.vn/payment/logo/Pay2Pay.jpg",
        "backGround": "https://static.pay2pay.vn.vn/payment/background/qr.jfif",
        "type": "QR"
      }
    ]
  }
}
```


V.4. INITIALIZE V1 - Static QRCode

API is used to initiate order payment transaction from merchant – Static QRCode. A static QR code is fixed and reused for multiple transactions. The payer enters the payment amount manually when scanning the code.

URL: https://{DOMAIN}/pgw-transaction-service/mch/api/v1.0/initialize

Method: POST

Request

#	Parameters	Data Type	Condition	Description
Header				
1	p-request-id	String	Mandatory	A unique code corresponding to a request sent from the merchant system
2	p-request-time	String	Mandatory	Time to request API (Unix time from 00:00:00 January 1, 1970). Format YYYYMMDDHHMMSS Note: It should be convert to UTC + 7
3	p-tenant	String	Mandatory	<TENANT INFO>. This information is sent along with username and password
4	Authorization	String	Mandatory	Accesstoken from login API
5	p-signature	String	Mandatory	The signature is generated by the RSA-2048 algorithm
Body				
1	currency	String	Mandatory	Currency Code
2	issuerId	String	Mandatory	Issuer Code: **PAY2PAY**
3	command	String	Mandatory	Command: **PAY**
4	paymentMethod	String	Mandatory	Payment method code
5	merchantData.orderId	String	Mandatory	Unique Order ID
6	orderDesc	String	Optional	Order Description
7	amount	Number	Mandatory	Amount to be paid

Response

#	Parameters	Data Type	Description
1	code	String	Result code
2	message	String	Message
3	data.qrInfo	String	QR information generated
4	data.qrMap	Object	Parsed QR information
5	data.providerCode	String	Code of the service provider handling the transaction
8	data.txnId	String	Transaction ID generated from Pay2Pay
9	data.bankInfo	Object	Bank information
10	data.bankInfo.binCode	String	Bank code
11	data.bankInfo.bankAccountName	String	Bank account name
12	data.bankInfo.bankAccountNumber	String	Bank account number

13	data.bankInfo.bankName	String	Bank name

EXAMPLE:

Request

```
curl --location 'https://{DOMAIN}/pgw-transaction-service/mch/api/v1.0/initialize' \
--header 'p-request-id: 4c632747-7df2-4386-a4f4-c59bfc65aeec' \
--header 'p-request-time: 20240523113751' \
--header 'p-tenant: MERCHANT-WEB' \
--header 'Authorization: Bearer
eyJhbGciOiJSUzI1NiIsInR5cCIgOiAiSldUIiwia2lkIiA6ICJFdFZjdmtDbUFYcWZWeU1TVjhhOThscnhUV1lJUl
JYRFprTER6UVZQekdBIn0.eyJleHAiOjE3MTY0Mzk1MTEsImlhdCI6MTcxNjQzODkxMSwianRpIjoiYzgyM2
YzODMtNjlhMC00NTQ2LTkwNGQtZDQwZDJhMTc0ZGFiIiwiaXNzIjoiaHR0cHM6Ly9wcGlkLnVhdC5wYXl
wYXkuY29tLnZuL3JlYWxtcy93YWxsZXQiLCJhdWQiOiJhY2NvdW50Iiwic3ViIjoiNjQ2YmQ5NGItMTI0MC
00NTNkLWI0NTYtZjY1OWRkN2MzOTFjIiwidHlwIjoiQmVhcmVyIiwiYXpwIjoicGF5cGF5LXdhbGxldCIsIn
Nlc3Npb25fc3RhdGUiOiIyODIwN2VjZS03NDQxLTQxOGYtYTIxMC1iODU0ODQ5YWFhMjciLCJhY3IiOiIx
IiwiYWxsb3dlZC1vcmlnaW5zIjpbIi8qIl0sInJlYWxtX2FjY2VzcyI6eyJyb2xlcyI6WyJvZmZsaW5lX2FjY2VzcyIs
InVtYV9hdXRob3JpemF0aW9uIiwiZGVmYXVsdC1yb2xlcy13YWxsZXQiXX0sInJlc291cmNlX2FjY2VzcyI6e
yJhY2NvdW50Ijp7InJvbGVzIjpbIm1hbmFnZS1hY2NvdW50IiwibWFuYWdlLWFjY291bnQtbGlua3MiLCJ2aW
V3LXByb2ZpbGUiXX19LCJzY29wZSI6IndhbGxldC1jbGllbnQtc2NvcGUgZW1haWwgcHJvZmlsZSIsInNpZCI
6IjI4MjA3ZWNlLTc0NDEtNDE4Zi1hMjEwLWI4NTQ4NDlhYWEyNyIsImVtYWlsX3ZlcmlmaWVkIjpmYWx
zZSwibWVyY2hhbnRJZCI6IlBQMDAwMDAwMSIsIm5hbWUiOiJkZWZhdWx0IGRlZmF1bHQiLCJncm91cH
MiOlsiL21hIiwiL21lcmNoYW50IiwiL29wcyIsIi9zYnYiLCIvd2FsbGV0Il0sInByZWZlcnJlZF91c2VybmFtZSI6Ij
A5ODI1Njk4ODciLCJnaXZlbl9uYW1lIjoiZGVmYXVsdCIsImZhbWlseV9uYW1lIjoiZGVmYXVsdCJ9.MKQw
U3YWT72UGksUS-
gC6r8DWaAYQyyKdRPUvp3QdQKLRKnXkuYo16PvSpftyEMMna2HtmGPHyoDFq1ospzr5ZIF6mrcDTe1Poc
Z9-_oiBqNStwb-6DZTcujC3pYjf1PH_dxfpNkCP43mI43mjTahYlKgxKxlF_f6FENmUDtyVmxW7S0gy7g92vR-
j2Yqa1LoIXbEqjYVRJUBaAFs98r6Zt34axuC5EccwAWVnCmDUNIq1xCW-
iNYaze26EK8ktTVo4XEQtHpE7qvzna8wVMQQ92TBWklFBmMsiQi7MQ4ZfAsjYrI5xFhZth2Ej-
il1ozP2OZtiqgyZl1Y2AWi9m2Q' \
--header 'p-signature:
Svdj2t7qS7k3TomsjsBmAUqyuhX0ECJ2WhC4KTU+5RVB5wlp9rmPTpz3E4bh96++2WBpZ4kZGSxXxaXCgkj
MWwgLt4LdSzwSgcbd5A5AjYuhIE6vU7CwxDAyWKu6gf4MALGV9y4WUPPSmAnTwLC/dzbkTNGs9Uimx
FG+5x6Fd0n0Yzag/jbCimecJ85FiAouh1F7mjiozelSiVsoWxn5L8Qb0ClwKyDGAyxpTMWW22v1z0sbVFBb702
3fHGiVuoWmP/mUv3zvAKl8/Cy0S+MhUri8Bvb4qM3oiohE3OSiO0C8P1Fz8eiKb0Ser3cSuur7nH9O2s817zhA
Xuo5sgpeg==' \
--header 'Content-Type: application/json' \
--data
'{"currency":"VND","issuerId":"PAY2PAY","command":"PAY","paymentMethod":"QRBANK","merchantData":
{"orderId":"664586dfe4a8aac0f0f8fef9","orderDesc":"Thanh toan don hang eCom","amount":"12050"}}' \
```

Response


```
{
  "code": "SUCCESS",
  "message": "Thành công",
  "data": {
    "qrInfo":
"00020101021238610010A0000007270131000697042201 17MBB55971988262877 0208QRIBFTTA5303704540
5120505802VN6219081520251027039491263048205",
    "qrMap": {
      "transactionCurrency": "704",
      "crc": "8205",
      "countryCode": "VN",
      "transactionAmount": "12050",
      "additionalInfo": {
        "purposeOfTransaction": "202510270394912"
      },
      "merchantInfo": {
        "accountInfo": {
          "binCode": "970422",
          "bankRefNumber": "MBB55971988262877"
        },
        "serviceType": "QRIBFTTA",
```

```
      "guid": "A000000727"
    },
    "type": "2",
    "payloadFormatIndicator": "01",
    "pointOfInitiationMethod": "12"
  },
  "bankInfo": {
    "binCode": "MB",
    "bankAccountName": "NGUYEN VAN AB",
    "bankAccountNumber": "MBB55971988262877",
    "bankName": "Ngân hàng TMCP Quân đội"
  },
  "providerCode": "MB",
  "txnId": "202510270394912"
}
}
```


V.5. INITIALIZE V2 - Dynamic QRCode

API is used to initiate order payment transaction from merchant – Dynamic QRCode. A dynamic QR code is generated for each payment request, containing order-specific details (e.g., amount, order ID). It can be scanned once by the customer to complete the transaction.

URL: https://{DOMAIN}/pgw-transaction-service/mch/api/v1.0/initialize

Method: POST

Request

#	Parameters	Data Type	Condition	Description
Header				
1	p-request-id	String	Mandatory	A unique code corresponding to a request sent from the merchant system
2	p-request-time	String	Mandatory	Time to request API (Unix time from 00:00:00 January 1, 1970). Format YYYYMMDDHHMMSS Note: It should be convert to UTC + 7
3	p-tenant	String	Mandatory	<TENANT INFO>. This information is sent along with username and password
4	Authorization	String	Mandatory	Accesstoken from login API
5	p-signature	String	Mandatory	The signature is generated by the RSA-2048 algorithm
Body				
1	currency	String	Mandatory	Currency Code
2	issuerId	String	Mandatory	IssuerId: PAYGATE
3	command	String	Mandatory	Command: PAY
4	paymentMethod	String	Mandatory	Payment method code
5	merchantData.orderId	String	Mandatory	Unique Order ID
6	orderDesc	String	Optional	Order Description
7	amount	Number	Mandatory	Amount to be paid

Response

#	Parameters	Data Type	Description
1	code	String	Result code
2	message	String	Message
3	data.qrInfo	String	QR information generated
4	data.qrMap	Object	Parsed QR information
5	data.bankInfo	Object	Bank information
6	data.respVACreated	Object	Information of the created virtual account
7	data.providerCode	String	Code of the service provider handling the transaction
8	data.txnId	String	Transaction ID generated from Pay2Pay
9	data.bankInfo	Object	Bank information
10	data.bankInfo.binCode	String	Bank code
11	data.bankInfo.bankAccountName	String	Bank account name
12	data.bankInfo.bankAccountNumber	String	Bank account number
13	data.bankInfo.bankName	String	Bank name

EXAMPLE:

Request

curl --location 'https://{DOMAIN}/pgw-transaction-service/mch/api/v1.0/initialize' \
--header 'p-request-id: e1522b07-796d-429d-b471-1c3edc254ed7' \
--header 'p-request-time: 20251216134246' \
--header 'p-tenant: MERCHANT-WEB' \
--header 'Authorization: Bearer
eyJhbGciOiJSUzI1NiIsInR5cCIgOiAiSldUIiwia2lkIiA6ICJFdFZjdmtDbUFYcWZWeU1TVjhhOThscnhUV1lJUlJ
YRFprTER6UVZQekdBIn0.eyJleHAiOjE3NjU4NjkxMDgsImlhdCI6MTc2NTg2NzMwOCwianRpIjoiNTliNjZjY
jEtZWMyZC00ZDRlLWI1OWMtYjY0Y2ZiZGExNzExIiwiaXNzIjoiaHR0cHM6Ly9wcGlkLnVhdC5wYXlwYX
kuY29tLnZuL3JlYWxtcy93YWxsZXQiLCJhdWQiOiJhY2NvdW50Iiwic3ViIjoiZjVhMjZkMWUtYjhhOC00YjA
wLTkxNDgtNTQ5NTE1YTA0NjkxIiwidHlwIjoiQmVhcmVyIiwiYXpwIjoicGF5cGF5LXdhbGxldCIsInNlc3Npb
25fc3RhdGUiOiJmYTAzNGNkOS05MGMzLTQwNTMtOTlkZC1jMGZmZGI1NDI2ZjkiLCJhY3IiOiIxIiwiYWx
sb3dlZC1vcmlnaW5zIjpbIi8qIl0sInJlYWxtX2FjY2VzcyI6eyJyb2xlcyI6WyJvZmZsaW5lX2FjY2VzcyIsInVtYV9
hdXRob3JpemF0aW9uIiwiZGVmYXVsdC1yb2xlcy13YWxsZXQiXX0sInJlc291cmNlX2FjY2VzcyI6eyJhY2Nv
dW50Ijp7InJvbGVzIjpbIm1hbmFnZS1hY2NvdW50IiwibWFuYWdlLWFjY291bnQtbGlua3MiLCJ2aWV3LXBy
b2ZpbGUiXX19LCJzY29wZSI6IndhbGxldC1jbGllbnQtc2NvcGUgZW1haWwgcHJvZmlsZSIsInNpZCI6ImZhM
DM0Y2Q5LTkwYzMtNDA1My05OWRkLWMwZmZkYjU0MjZmOSIsImVtYWlsX3ZlcmlmaWVkIjp0cnVlLC
JtZXJjaGFudElkIjoiUFAwMDAwMjQ5IiwiYXBwSWQiOlsiaW50LWFwaS51YXQucGF5MnBheS52biIsImludC
1hcGkudWF0LnBheW1lbnRob3QuY29tIiwidWF0LWFwaS5wYXkycGF5LnZuIl0sImdyb3VwcyI6WyIvbWVyY
2hhbnQiLCIvb3BzIl0sInByZWZlcnJlZF91c2VybmFtZSI6ImhvYW5uZ3V5ZW5odXUiLCJlbWFpbCI6ImhvYW
5uZ3V5ZW5odXVAcGF5MnBheS52biJ9.fib7pwteRwG5db6zRIBBUq8vltEN9x5KDnD9kO03Eweo2AhaoG-
a6Igl8mgmWXcsV4zgzMBsmL6dOkRNom1dG9GSdlbGl38yujqY1swiqSkqkW1aoUQZivXlCBEkAxJ7wk6qJ8
w-dt0S2Js8avji-Q3MtOoIzp4J6fuIvHPddtnSVt6Xtcj8RFzA8rLAF-
_hdObt9AyXc8aukuGz83wfTy9A14lJTuqXREi2OeU-
UdfMZ2YpeiYRnRzdyaaxcN7Ie96R4n5evo21At1zfZtouROXWFQ0eR70qJ3gGPUnqzflraCmm0jfti9h0f_eDeVP
gALEAZELKZuA5ulPqKxzoA' \
--header 'verification: 97a38363521c48c8a6383deea132c227' \
--header 'p-signature:
KcD9cs9MQMgN7MqeMzkQsnzAX0nI1ZB3OzfLoMermTuCkL1z4Og6S6BdjKXZ/iKoPdH02JZ8xiAJMmRKO

HMNU8McjtiSCocYIpLh/MgUDukrsaht81qa0nC6PguHY2lNW+Fmu/Zn8XW/TwHiymX+8vPlRduztIqAN7B7rmHA716031WCGr3rbRGMcUYfhDmWdybnRkNlb1swK+LzloctcD7QuyqFlft63bAozw6eBaASyQ18ujEs/DqdKiEnwAmDfgZhH6hXem18/Ulxhj+g7EaaqEH2zaOrCFYMymKtl7amE2eo7s/+48hFuCmKm2fK+xvjiUgiJw03Jt6o61ZZDg==' \
--header 'Content-Type: application/json' \
--data
'{"currency":"VND","issuerId":"PAYGATE","command":"PAY","paymentMethod":"QRBANK","merchantData":{"orderId":"664586dfe4a8aac0f0f8lal9","orderDesc":"Thanh toan don hang eCom","amount":"500000"}}' \

Response

{
"code": "SUCCESS",
"message": "Thành công",
"data": {
"qrInfo":
"00020101021238610010A0000007270131000697042201 17MBB55679376395013 0208QRIBFTTA53037045406 5000005802VN6219081520251216040445463049a97",
"qrMap": {
"transactionCurrency": "704",
"crc": "9a97",
"countryCode": "VN",
"transactionAmount": "500000",
"additionalInfo": {
"purposeOfTransaction": "202512160404454"
},
"merchantInfo": {
"accountInfo": {
"binCode": "970422",
"bankRefNumber": "MBB55679376395013"
},
"serviceType": "QRIBFTTA",
"guid": "A000000727"
},
"type": "2",
"payloadFormatIndicator": "01",
"pointOfInitiationMethod": "12"
},
"bankInfo": {
"binCode": "MB",
"bankAccountName": "VA TOYS",
"bankAccountNumber": "MBB55679376395013",
"bankName": "Ngân hàng TMCP Quân đội"
},
"respVACreated": {
"merchantCode": "PP0000249",
"bankRefNumber": "MBB55679376395013",
"bankRefName": "VA TOYS",
"bankId": "MB",
"requestAmount": null,
"status": "ACTIVE",
"qrUsageType": null,
"provider": "MB",
"orderExpireTime": 20251216143246,
"accountType": "ONE_TIME"
},
"providerCode": "MB",
"txnId": "202512160404454"
}
}

V.6. INQUIRY V1

API to get status off transaction by Order ID

URL: https://{DOMAIN}/pgw-transaction-service/mch/api/v1.0/inquiry

Method: POST

Request

#	Parameters	Data Type	Condition	Description
Header				
1	p-request-id	String	Mandatory	A unique code corresponding to a request sent from the merchant system
2	p-request-time	String	Mandatory	Time to request API (Unix time from 00:00:00 January 1, 1970). Format YYYYMMDDHHMMSS Note: It should be convert to UTC + 7
3	p-tenant	String	Mandatory	<TENANT INFO>. This information is sent along with username and password
4	Authorization	String	Mandatory	Accesstoken from login API
5	p-signature	String	Mandatory	The signature is generated by the RSA-2048 algorithm
				-
Body				
1	fromDate	Number	Mandatory	From Date. Format: YYYYMMDD
2	toDate	Number	Mandatory	To Date Format: YYYYMMDD
3	orderId	Number	Mandatory	Merchant Order ID

Response

#	Parameters	Data Type	Description
1	code	String	Result code
2	message	String	Message
3	data.transId	String	Pay2Pay Transaction ID
4	data.code	String	Status of Transaction (INIT,SUCCESS,FAIL,PENDING,SUSPECT,CANCELED)

EXAMPLE:

Request

```
curl --location 'https:// {DOMAIN}/pgw-transaction-service/mch/api/v1.0/inquiry' \
--header 'p-request-id: a32fac8e-439d-4858-b7f8-63b106fc3538' \
```

--header 'p-request-time: 20240521175911' \
--header 'p-tenant: MERCHANT-WEB' \
--header 'Authorization: Bearer
eyJhbGciOiJSUzI1NiIsInR5cCIgOiAiSldUIiwia2lkIiA6ICJFdFZjdmtDbUFYcWZWeU1TVjhhOThscnhUV1lJUl
JYRFprTER6UVZQekdBIn0.eyJleHAiOjE3MTYyODk3MTQsImlhdCI6MTcxNjI4OTExNCwianRpIjoiY2I0ZDk
zMzMtMDc3ZS00YzY2LWE0NGMtZjg1ZjMwMTIyNjc4IiwiaXNzIjoiaHR0cHM6Ly9wcGlkLnVhdC5wYXlw
YXkuY29tLnZuL3JlYWxtcy93YWxsZXQiLCJhdWQiOiJhY2NvdW50Iiwic3ViIjoiOWVhMmY3OGItNzg2Zi00
MTZlLWFlZGEtYWEyMGUzNjY0ZGQ3IiwidHlwIjoiQmVhcmVyIiwiYXpwIjoicGF5cGF5LXdhbGxldCIsInNl
c3Npb25fc3RhdGUiOiJjNzU5Njg2My0wNGMzLTQzMGItOTJmMi1hMTcxNzhlNzAyZGMiLCJhY3IiOiIxIiwi
YWxsb3dlZC1vcmlnaW5zIjpbIi8qIl0sInJlYWxtX2FjY2VzcyI6eyJyb2xlcyI6WyJvZmZsaW5lX2FjY2VzcyIsInVt
YV9hdXRob3JpemF0aW9uIiwiZGVmYXVsdC1yb2xlcy13YWxsZXQiXX0sInJlc291cmNlX2FjY2VzcyI6eyJhY
2NvdW50Ijp7InJvbGVzIjpbIm1hbmFnZS1hY2NvdW50IiwibWFuYWdlLWFjY291bnQtbGlua3MiLCJ2aWV3L
XByb2ZpbGUiXX19LCJzY29wZSI6IndhbGxldC1jbGllbnQtc2NvcGUgZW1haWwgcHJvZmlsZSIsInNpZCI6Im
M3NTk2ODYzLTA0YzMtNDMwYi05MmYyLWExNzE3OGU3MDJkYyIsImVtYWlsX3ZlcmlmaWVkIjpmYW
xzZSwibWVyY2hhbnRJZCI6IlBQMDAwMDA3MCIsIm5hbWUiOiJkZWZhdWx0IGRlZmF1bHQiLCJncm91cH
MiOlsiL21hIiwiL21lcmNoYW50IiwiL29wcyJdLCJwcmVmZXJyZWRfdXNlcm5hbWUiOiIwNzgzMTgwODc4Ii
wiZ2l2ZW5fbmFtZSI6ImRlZmF1bHQiLCJmYW1pbHlfbmFtZSI6ImRlZmF1bHQifQ.LkP5vTIPA7P8nbYTmIt0
htIWVty97NoANp0_kyFIJoCMNjAzciPCrQd08OpHoyqkRAbEJ-EXTPY-
5q0DPr3x10WfzTECnxSJyCMUTvrLAcRGzmlteF-j4oCPjVzRFqKPaOdZ0S9f-
LQzmmwraslXGme3AqU1y2Vlj49h7h1SjsNfEUbxQ6iI8CGMEiFq4yVHCBisAjvBINVGfr_EsL7_7i_G6AS25O
o1h7fDl4JXVbNQcvq8_m040kPHBVeNv1R8_qr75pLg9QjUl1xBZq2XoJxamgvUpe1ps8Z9l2JXV3kc6DwUeBD
YJdVhucnB0lTfJp4QjqAyQjX2mU50b51lIg' \
--header 'p-signature:
ZCabIfkJOWU25zIgMnFVP180NbgeJmkF0MClVBTBAtQujJm8qHeEKosj53ummXJ96M39Lszwhqcz8xtGJeA
AvhOsHuga9i/20CHJHYZ/L+n21+ZyYUApeiW+peLSwY/DMflXuFgeafgfGmabzgr2Ajk30gI86TUdW6hlqwexh
Op1TbQ8OvFXxE0GDnPMa3br6JEdBfJQ21w+MHDFCnWr9SXFA/rOMTuHY7RPmB2RaXhWz5JMet/NaBbd
LJddo6atuKpYAWp+wl91DYNJ1nHrfMJgeoITcxJxJkptwlAUuSqlWl88yHpL+ybK6lz781u2MDQ7vhkB1t8I0Sn
AYA1usA==' \
--header 'Content-Type: application/json' \
--data '{"fromDate":"20240501","toDate":"20240531","orderId":"66419d4bbac3351da1d98668"}' \

Response

{
 "code": "SUCCESS",
 "message": "Thành công.",
 "data": {
  "transId": "202405130004096",
  "code": "SUCCESS"
 }
}

V.7. INQUIRY V2

API to get status off transaction by Order ID
URL: https://{DOMAIN}/pgw-transaction-service/mch/api/v2.0/inquiry
Method: POST
Request

#	Parameters	Data Type	Condition	Description
Header				
1	p-request-id	String	Mandatory	A unique code corresponding to a request sent from the merchant system
2	p-request-time	String	Mandatory	Time to request API (Unix time from 00:00:00 January 1, 1970). Format YYYYMMDDHHMMSS Note: It should be convert to UTC + 7
3	p-tenant	String	Mandatory	<TENANT INFO>. This information is sent along with username and password
4	Authorization	String	Mandatory	Accesstoken from login API

5	p-signature	String	Mandatory	The signature is generated by the RSA-2048 algorithm
				-
Body				
1	orderId	Number	Mandatory	Merchant Order ID

Response

#	Parameter	Data Type	Description
1	merchantId	String	Merchant ID
2	orderId	String	Purchase Order ID generated by merchant (unique)
3	txnId	String	Pay2Pay Transaction ID
4	amount	Number	Amount of purchase order
5	status	String	Status of transaction (INIT, SUCCESS, FAIL, PENDING, SUSPECT, CANCELED)
6	code	String	Transaction code
7	message	String	Description of transaction status
8	txnDate	String	Transaction time
9	secretSenderBankRefNumber	String	Base64 encoded string of sender's bank account number
10	senderBankRefName	String	Bank account name of sender
11	senderBankId	String	Bank Id of sender
12	senderBankName	String	Bank Name of sender
13	senderBinCode	String	Bank Bin Code of sender
14	receiverBankId	String	Bank Id of receiver
15	receiverBankRefName	String	Bank Name of receiver
16	receiverBankRefNumber	String	Bank Bin Code of receiver

17	content	String	Content of transaction

EXAMPLE:

Request

```
curl --location 'https://uat-api.pay2pay.vn/pgw-transaction-service/mch/api/v2.0/inquiry' \
--header 'p-request-id: 1e7523c6-7f62-4cab-9bac-8972fcf2839b' \
--header 'p-request-time: 20251219143405' \
--header 'p-tenant: MERCHANT-WEB' \
--header 'Authorization: Bearer
eyJhbGciOiJSUzI1NiIsInR5cCIgOiAiSldUIiwia2lkIiA6ICJFdFZjdmtDbUFYcWZWeU1TVjhhOThscnhUV1lJUlJ
YRFprTER6UVZQekdBIn0.eyJleHAiOjE3NjYxMzEzODYsImlhdCI6MTc2NjEyOTU4NiwianRpIjoiNzdhYjVj
MmYtNTRkZi00ZGI1LWI2OWMtYjU1MDg1NzYxNWM1IiwiaXNzIjoiaHR0cHM6Ly9wcGlkLnVhdC5wYXl
wYXkuY29tLnZuL3JlYWxtcy93YWxsZXQiLCJhdWQiOiJhY2NvdW50Iiwic3ViIjoiZjVhMjZkMWUtYjhhOC0
0YjAwLTkxNDgtNTQ5NTE1YTA0NjkxIiwidHlwIjoiQmVhcmVyIiwiYXpwIjoicGF5cGF5LXdhbGxldCIsInNlc3
Npb25fc3RhdGUiOiI2ZjVjYTJiZC05NzUwLTQyZWEtODFlNi1lY2EwOWIyMTIwOGYiLCJhY3IiOiIxIiwiYW
xsb3dlZC1vcmlnaW5zIjpbIi8qIl0sInJlYWxtX2FjY2VzcyI6eyJyb2xlcyI6WyJvZmZsaW5lX2FjY2VzcyIsInVtYV
9hdXRob3JpemF0aW9uIiwiZGVmYXVsdC1yb2xlcy13YWxsZXQiXX0sInJlc291cmNlX2FjY2VzcyI6eyJhY2N
vdW50Ijp7InJvbGVzIjpbIm1hbmFnZS1hY2NvdW50IiwibWFuYWdlLWFjY291bnQtbGlua3MiLCJ2aWV3LXB
yb2ZpbGUiXX19LCJzY29wZSI6IndhbGxldC1jbGllbnQtc2NvcGUgZW1haWwgcHJvZmlsZSIsInNpZCI6IjZmN
WNhMmJkLTk3NTAtNDJlYS04MWU2LWVjYTA5YjIxMjA4ZiIsImVtYWlsX3ZlcmlmaWVkIjp0cnVlLCJtZXJ
jaGFudElkIjoiUFAwMDAwMjQ5IiwiYXBwSWQiOlsiaW50LWFwaS51YXQucGF5MnBheS52biIsImludC1hcGk
udWF0LnBheW1lbnRob3QuY29tIiwiaW50LWFwaS5kZXYucGF5cGF5LmNvbS52biIsInVhdC1hcGkucGF5Mn
BheS52biJdLCJncm91cHMiOlsiL21lcmNoYW50IiwiL29wcyJdLCJwcmVmZXJyZWRfdXNlcm5hbWUiOiJob2
Fubmd1eWVuaHV1IiwiZW1haWwiOiJob2Fubmd1eWVuaHV1QHBheTJwYXkudm4ifQ.ThP5b-
_7pR4R7bUxJoJ6RSo1YBwsIcroyIB9xk90xvCzOilEOgYEPaXRduBeN0cfFI-
vdW1y5tIAMQovhdvX2b2FfjvPT584sXYG2hKGjczVMVTocduFlhB98rtHUmL6Sftbqw3OZGcCb7d3EwNAkP
8dUqD7-eQ1mlm9XjRK39iVC_oEfMAC5-D_dNAudDB7I4gc72Fgndfj1KL-
lFnN47aY2ucMMbukvBkxgcRjfpkWPA-gwE-PxKSaJXeQ5j-wCa5apgI-WsF4WxJc1wZxZOgTohiSGA-
xWvdSAHiaigXUeSPWCtKt0UuvENl8ftNKXAY2lkEtFxtX0VjxsBZoIA' \
--header 'verification: 32c1cdf335f3491cbb5ce2edc5b0ec25' \
--header 'p-signature:
Sb01kRpX43OXe3Kc5Y6VVROYgQaDKXJTzUVecjn9OiHsPfu8R/KhH56ztDq+dQg3EoC55ceYy3lU5Bi09f2O
uc8JGZ4qHBuTheMAIqOjDp0kDqPKyxXgwDdOXOhJTgj3YDQjRkZQkgfziubgIUpJKNV/5jXSSe2hpecOdU7
KkL9ErRj3obKrNqR37hmt7axXdr5yhx+F6fwShcuQaf3uwk2ZAOH0AmfsXtyzSEVtGlkuN0srE0klb1eaeo1Bi6F
5CDIO23oylBPDw/uzaiFrlEupDwrnYH08mEBPi+FnYnsRzQsYNW22toPfodK0eX9no4h/dQYk1AU0Z6ToE3/q
KQ==' \
--header 'Content-Type: application/json' \
--data '{"orderId":"f04fa475c201440999f68161ec054395"}' \
```

Response


```
{
  "code": "SUCCESS",
  "message": "Thành công.",
  "data": {
    "merchantId": "PP0000042",
    "orderId": "4542a9b3e08d766b0532f9a8",
    "txnId": "202512190404595",
    "amount": "100001",
    "code": "SUCCESS",
    "status": "SUCCESS",
    "message": "Giao dịch thành công",
    "txnDate": "20251219103418",
    "secretSenderBankRefNumber": "MDk2ODA5Mjk2Nw==",
    "senderBankRefName": "AUTOMATION TEST GJE",
    "senderBankId": "MB",
    "senderBankName": "MBBank",
    "senderBinCode": "970422",
    "receiverBankId": "MB",
    "receiverBankRefName": "Mb check",
    "receiverBankRefNumber": "MBB55464374293555"
  }
}
```

V.8. REFUND

API to request refund one transaction

URL: https://{DOMAIN}/pgw-transaction-service/mch/api/v1.0/init_refund

Method: POST

Request

#	Parameters	Data Type	Condition	Description
Header				
1	p-request-id	String	Mandatory	A unique code corresponding to a request sent from the merchant system
2	p-request-time	String	Mandatory	Time to request API (Unix time from 00:00:00 January 1, 1970). Format YYYYMMDDHHMMSS Note: It should be convert to UTC + 7
3	p-tenant	String	Mandatory	<TENANT INFO>. This information is sent along with username and password
4	Authorization	String	Mandatory	Accesstoken from login API
5	p-signature	String	Mandatory	The signature is generated by the RSA-2048 algorithm
Body				
1	txnId	Number	Mandatory	Pay2Pay Transaction ID
2	amount	Number	Mandatory	Transaction Amount
3	description	String	Optional	Description

Response

#	Parameters	Data Type	Description
1	code	String	Result code
2	message	String	Message

EXAMPLE:

Request

```
curl --location 'https://{DOMAIN}/pgw-transaction-service/mch/api/v1.0/init_refund' \
--header 'p-request-id: 7215ad01-4ae7-4fd3-8ef5-00658f37e924' \
--header 'p-request-time: 20240522094049' \
--header 'p-tenant: MERCHANT-WEB' \
--header 'Authorization: Bearer
eyJhbGciOiJSUzI1NiIsInR5cCIgOiAiSldUIiwia2lkIiA6ICJFdFZjdmtDbUFYcWZWeU1TVjhhOThscnhUV1lJUl
JYRFprTER6UVZQekdBIn0.eyJleHAiOjE3MTYzNDYwNzksImlhdCI6MTcxNjM0NTQ3OSwianRpIjoiMzlkN
mQ0YzQtNTUzZC00YTMwLWE1MzUtMDE1Y2JhMDJkMmQ4IiwiaXNzIjoiaHR0cHM6Ly9wcGlkLnVhdC5
wYXlwYXkuY29tLnZuL3JlYWxtcy93YWxsZXQiLCJhdWQiOiJhY2NvdW50Iiwic3ViIjoiNjc0NTgzYzMtOWE
1MC00OTMwLTk3N2UtM2I0YmE1MmE4ZDNiIiwidHlwIjoiQmVhcmVyIiwiYXpwIjoicGF5cGF5LXdhbGxld
```

CIsInNlc3Npb25fc3RhdGUiOiI5MmZkYWFmMy1lNTI1LTRhY2EtYWVhOS1lOGMxYjk4NjQ3MDUiLCJhY3IiOiIxIiwiYWxsb3dlZC1vcmlnaW5zIjpbIi8qIl0sInJlYWxtX2FjY2VzcyI6eyJyb2xlcyI6WyJvZmZsaW5lX2FjY2VzcyIsInVtYV9hdXRob3JpemF0aW9uIiwiZGVmYXVsdC1yb2xlcy13YWxsZXQiXX0sInJlc291cmNlX2FjY2VzcyI6eyJhY2NvdW50Ijp7InJvbGVzIjpbIm1hbmFnZS1hY2NvdW50IiwibWFuYWdlLWFjY291bnQtbGlua3MiLCJ2aWV3LXByb2ZpbGUiXX19LCJzY29wZSI6IndhbGxldC1jbGllbnQtc2NvcGUgZW1haWwgcHJvZmlsZSIsInNpZCI6IjkyZmRhYWYzLWU1MjUtNGFjYS1hZWE5LWU4YzFiOTg2NDcwNSIsImVtYWlsX3ZlcmlmaWVkIjpmYWxzZSwibWVyY2hhbnRJZCI6IlBQMDAwMDA3MCIsImdyb3VwcyI6WyIvbWVyY2hhbnQiLCIvb3BzIl0sInByZWZlcnJlZF91c2VybmFtZSI6InR1YW5sZSJ9.m6ZRxQiuEGUybGd0Yg9qL00-9wMk12XjgYtp6HRCorLVRA8b4F_wpN1RFpTslGAb-hb5af7yQirB0gDP90hYCiNgOK4M_s9ZTGMXOMK5cDKrdL5ZiCKz6ak1lPhGw5jyyOn1s5qcYAgRDqUuwzXtjN3331wc3FHYJPA4NviVAFyWlFUaEnxSBzGclW0vGAwNG4TgJ-kQkt2uuUMSaYhB9d5VEtzpvz5ok-GmofcWLf6lnt5GdcrYjwJnh89kcJK4gN4ouKdFp3lokwWbvgIp9Bh4siUFv0K4RfQGlZAXuY0we1RulW7eP2YtmPxxZkagKUM7RCF-CH7BJcJ2sruSdg' \
--header 'p-signature:
W13oljGxrWOoGZON8DeRjoP7/9drsFyoJOrjQweNWQv5JFpkK8FML4Y2uc0KCM8wunZjDNZc1Wdyoa4Hkzmjd8DyECbA9qTSam5t2kXkWmYGmkZKddVpUtyiaC+NQAzU+oyEAo9ujhlpcLWIXiPGN0dkOoXMifXAeyrcoLIjgHScAJRjSSPVydXWpuxYYH34thlK+YO0L9u+698P6y7TfX83jM/3tU74v6XOo6XCUeDdch2UTVJOEMY9E0BAAXFA+Th0AIFBSmht3///KCzibC7smebhIClnv1cluPqrIZvUlRFc2DMazGIq4lP9BOpCmr8RQvpFg6Zd3LklX+t9A==' \
--header 'Content-Type: application/json' \
--data '{"txn":[{"txnId":202405140004160,"amount":108000,"description":"hoan tien"}]}' \

Response

```
{
  "code": "SUCCESS",
  "message": "Thành công.",
}
```

V.9. IPN – Instant Payment Notification

Merchant will receive results from Pay2Pay for the transaction flow via IPN (POST method). This result is captured in requestBody.

Method: POST

Request Header

#	Parameter	Data Type	Description
1	P-API-KEY	String	Merchant's Api key
2	P-SIGNATURE	String	Signature generate from whole request body & secret key

Request Body

#	Parameter	Data Type	Description
1	merchantId	String	Merchant ID
2	orderId	String	Purchase Order ID generated by merchant (unique)
3	txnId	String	Pay2Pay Transaction ID
4	amount	Number	Amount of purchase order
5	status	String	Status code (SUCCESS/FAIL/PROCESSING)

6	code	String	Transaction code, Please refer to VI. List of API response code
7	message	String	Description of transaction status
8	txnDate	String	Transaction time
9	secretSenderBankRefNumber	String	Base64 encoded string of sender's bank account number
10	senderBankRefName	String	Bank account name of sender
11	senderBankId	String	Bank Id of sender
12	senderBankName	String	Bank Name of sender
13	senderBinCode	String	Bank Bin Code of sender
14	receiverBankId	String	Bank Id of receiver
15	receiverBankRefName	String	Bank Name of receiver
16	receiverBankRefNumber	String	Bank Bin Code of receiver
17	content	String	Content of transaction

Steps to Create Signature:

Step 1: convert request body to JSON string

{"merchantId":" PP0000042","orderId":"
be4e93a4f8af8661cf758604","txnId":"202511070395223","amount":400000,"code":" SUCCESS ","message":"Transaction is
successful.","txnDate":"20251107163546","secretSenderBankRefNumber":"MDc3ODc2MTE0OA==","senderBankRefName":"AUTOMATION TEST
YZL","senderBankId":"MB","senderBankName":"MBBank","senderBinCode":"970422","content":"test","receiverBankId":"MB","receiverBankRefName":"NGUYEN VAN
AB","receiverBankRefNumber":"MBB55971988262877","status":"SUCCESS"}

Step 2: append secretKey value to the result of **Step 1**

{"merchantId":" PP0000042","orderId":"
be4e93a4f8af8661cf758604","txnId":"202511070395223","amount":400000,"code":" SUCCESS ","message":"Transaction is
successful.","txnDate":"20251107163546","secretSenderBankRefNumber":"MDc3ODc2MTE0OA==","senderBankRefName":"AUTOMATION TEST
YZL","senderBankId":"MB","senderBankName":"MBBank","senderBinCode":"970422","content":"test","receiverBankId":"MB","receiverBankRefName":"NGUYEN VAN
AB","receiverBankRefNumber":"MBB55971988262877","status":"SUCCESS"} <secretKey value>

Step 3: Apply the SHA-256 hash algorithm to string value of Step 2

Step 2's Value:
{"merchantId":" PP0000042","orderId":"
be4e93a4f8af8661cf758604","txnId":"202511070395223","amount":400000,"code":" SUCCESS ","message":"Transaction is
successful.","txnDate":"20251107163546","secretSenderBankRefNumber":"MDc3ODc2MTE0OA==","senderBankRefName":"AUTOMATION TEST
YZL","senderBankId":"MB","senderBankName":"MBBank","senderBinCode":"970422","content":"test","receiverBankId":

"MB","receiverBankRefName":"NGUYEN VAN AB","receiverBankRefNumber":"MBB55971988262877","status":"SUCCESS"} MyScretKey

Signature value: use this value to verify with header signature
qi2bkXcxhiDnZMVFXn6z64wCSagpD9ujquDGRSzemg0=

Refer link to check & get snippet code:

https://static.pay2pay.vn/tools/ipn_validate

Example:

Data Sample

Request Header:

P-API-KEY : 123456

P-SIGNATURE: qi2bkXcxhiDnZMVFXn6z64wCSagpD9ujquDGRSzemg0=

Request Body:


```
{
 "merchantId": " PP0000042",
 "orderId": " be4e93a4f8af8661cf758604",
 "txnId": "202511070395223",
 "amount": 400000,
 "code": " SUCCESS ",
 "message": "Transaction is successful.",
 "txnDate": "20251107163546",
 "secretSenderBankRefNumber": "MDc3ODc2MTE0OA==",
 "senderBankRefName": "AUTOMATION TEST YZL",
 "senderBankId": "MB",
 "senderBankName": "MBBank",
 "senderBinCode": "970422",
 "content": "test",
 "receiverBankId": "MB",
 "receiverBankRefName": "NGUYEN VAN AB",
 "receiverBankRefNumber": "MBB55971988262877",
 "status": "SUCCESS"
}
```


VI. LIST OF API STATUS CODES

Status Codes indicate whether a transaction has succeeded, failed, or is still being processed. Merchants use these codes to decide how to handle each transaction, such as confirming success, waiting for completion, or retrying when necessary.

No	Status	Description
1	SUCCESS	The transaction has been completed successfully. The final result is confirmed, and no further action is required. Applicable to both collection and payout.
2	PROCESSING	The transaction has been received and is currently being processed by the bank or payment provider. The final status will be updated later through callback.
3	FAIL	The transaction could not be completed. This may be due to incorrect information, insufficient balance, exceeded limits, timeout, or external bank rejection. Final state—no further processing will occur.

4	INIT	The transaction has been initialized and recorded in the system, but no processing has started yet. This status indicates the initial creation stage before being sent for processing.
5	CANCELED	The transaction has been canceled intentionally by the user or the system before completion. No further processing will occur, and no funds will be deducted or transferred.
6	SUSPECT	The transaction has been flagged as suspicious due to abnormal behavior, potential fraud, or risk control rules. Further verification or manual review is required before determining the final result.

VII. LIST OF API RESPONSE CODES

No	Code	Message
1	SUCCESS	Success code
2	ERR_BANK_SYSTEM	Error system
3	ERR_BANK_TIMEOUT	Error system
4	ERR_BANK_ACCOUNT_IS_NOT_CORRECT	Account name or number is not correct
5	ERR_BANK_REQUEST_ID_ALREADY_EXIST	The request id already exists
6	ERR_BANK_INVALID_PARAMETER	Invalid parameter
7	ERR_BANK_BANK_INFORMATION_NOT_FOUND	Bank information not found
8	ERR_BANK_TRANSACTION_FAIL	Transaction failed
9	ERR_BANK_NOT_RECEIVE_RESPONSE_FROM_BANK	Did not receive response from bank to transfer money
10	ERR_BANK_CURRENT_BANK_DISCONTINUITY	The current beneficiary bank is discontinuity
11	ERR_BANK_ISSUING_BANK_REFUSES_PROCESS_REQUEST	Issuing bank refuses to process this request
12	ERR_BANK_PAYING_BANK_REFUSES_PROCESS_REQUEST	The paying bank refuses to process this request
13	ERR_BANK_MERCHANT_ACTIVE_BANK_VA_INFO_DOES_NOT_EXIST	Merchant's account haven't declared
14	ERR_BANK_TRANSACTION_DOES_NOT_EXIST	Transaction does not exist
15	ERR_BANK_TRANSACTION_STATUS_INVALID	The status of the transaction is invalid
16	ERR_SYSTEM	system error.
17	ERR_TRANSACTION_MERCHANT_INVALID	invalid merchant.
18	ERR_DATA_INVALID	invalid data.
19	ERR_TOTALAMOUNT_INVALID	Invalid Total amount.
20	ERR_TRANSACTION_METHOD_INVALID	Invalid payment method.
21	ERR_TRANSACTION_ISSUER_INVALID	Invalid issuer.
22	ERR_TRANSACTION_CURRENCY_INVALID	Invalid currency amount.
23	ERR_TRANSACTION_INIT_DATA_NOT_FOUND	Init data not found.
24	ERR_TRANSACTION_NOT_FOUND	Transaction not found.
25	MERCHANT_INVALID	invalid merchant.
26	ERR_MERCHANT_DOES_NOT_EXIST	Merchant does not exist
27	ERR_MERCHANT_FEE_INVALID	Invalid transaction fee
28	ERR_MERCHANT_TRANSACTION_INVALID	Invalid transaction
29	ERR_MERCHANT_CORE_SUSPECT	System error. Please try again in a few minutes
30	ERR_MERCHANT_CORE_TIMEOUT	System error. Please try again in a few minutes
31	ERR_MERCHANT_BANK_TIMEOUT	System error. Please try again in a few minutes

32	ERR_MERCHANT_BANK_SUSPECT	System error. Please try again in a few minutes

VIII. CODE SAMPLE

1. PHP

```
function computeSha256HashBase64($inputString) {
   // Compute the SHA-256 hash of the input string
   $hash = hash('sha256', $inputString);
   // Encode the resulting hash in Base64
   $base64Hash = base64_encode($hash);
   return $base64Hash;
}
```

2. C#

```
public static string ComputeSha256HashBase64(string inputString)
   {
      byte[] inputBytes = Encoding.UTF8.GetBytes(inputString);
      // Create a SHA256 instance
      using (SHA256 sha256 = SHA256.Create())
      {
         byte[] hashBytes = sha256.ComputeHash(inputBytes);
         string base64Hash = Convert.ToBase64String(hashBytes);
         return base64Hash;
      }
   }
```

3. ReactJs

```
import CryptoJS from 'crypto-js';
function computeSha256HashBase64(inputString) {
   // Compute SHA-256 hash
   const hash = CryptoJS.SHA256(inputString);

   // Convert the hash to a Base64 string
   const base64Hash = CryptoJS.enc.Base64.stringify(hash);

   return base64Hash;
}
```

4. Java

```
public static String computeSha256HashBase64(String inputString) {
      try {
         MessageDigest digest = MessageDigest.getInstance("SHA-256");
         byte[] hashBytes = digest.digest(inputString.getBytes(StandardCharsets.UTF_8));
         String base64Hash = Base64.getEncoder().encodeToString(hashBytes);
         return base64Hash;
      } catch (NoSuchAlgorithmException e) {
         throw new RuntimeException(e);
      }
   }
```